E-COMMERCE

Profitability Analysis Report

Prepared from: E-Commerce_profitability_analysis.sql
Analysis Period: 2024 – 2025
Data Sources: orders.csv | products.csv | marketing_spend.csv

2,000	$277.9K	20.3%	16.2x
Total Orders	Gross Revenue	Avg Profit Margin	Overall ROAS

1. Data Quality & Validation

Before analysis, all three tables were validated for integrity issues. The SQL script performs the following checks:

Checks performed

- Duplicate primary keys in orders (order_id) and products (product_id)
- Composite primary key validation for marketing_spend (month, platform)
- NULL checks across all key identifier columns
- Cost reconciliation: product_cost + shipping_cost + platform_fee + transaction_fee = total_costs
- Negative value checks on revenue, cost, and profit columns
- Discount amount never exceeds gross revenue
- Distinct value audits for channel, payment_method, and region

Result	**No data quality issues found. All cost components reconcile correctly to total_costs. Zero duplicate keys or NULL identifiers detected.**

2. Profit Margin by Product Category (Q1)

Results

Category	Net Revenue	Total Profit	Margin %	Avg Shipping	Avg Disc %	Return Rate
Electronics	$44,886	$13,973	**31.1%**	$26.74	7.49%	8.61%
Toys	$33,596	$8,786	26.2%	$26.23	8.52%	7.00%
Home & Kitchen	$26,360	$6,689	25.4%	$24.94	7.28%	6.00%
Food & Beverage	$30,663	$7,592	24.8%	$26.08	7.13%	5.67%
Sports	$32,330	$7,597	23.5%	$23.68	8.00%	7.19%
Clothing	$31,383	$6,272	20.0%	$25.41	7.32%	8.19%
Beauty	$18,254	$3,175	17.4%	$25.23	7.83%	5.85%
Books	$18,847	$2,250	**11.9%**	$26.20	7.45%	8.37%

Key findings

- Electronics leads with a **31.1%** margin, driven by high average selling prices absorbing fixed shipping costs.
- Books have the worst margin at **11.9%** — high return rate (8.4%) combined with high avg shipping cost ($26.20) on low-value items is the primary drag.
- Shipping cost is the most variable driver — **Beauty and Books both suffer from high shipping per-order relative to their selling prices.**
- Food & Beverage has the lowest return rate (5.7%) which partially offsets moderate margins.

CEO Insight	**To improve Books and Beauty margins, focus on shipping rate negotiation or minimum order thresholds — product cost reduction alone will not solve the problem.**

3. Profitability by Sales Channel (Q2)

Results

Channel	Orders	Net Revenue	Total Profit	Profit/Order	Margin %
Mobile App	589	$71,893	$21,393	**$36.32**	29.8%
Website	795	$92,991	$25,118	$31.60	27.0%
Social Commerce	197	$21,929	$3,371	$17.11	15.4%
Marketplace	419	$49,505	$6,452	**$15.40**	13.0%

Key findings

- Mobile App is the most profitable channel at **$36.32 profit per order** and 29.8% margin — zero platform fees.
- Website is the highest revenue channel **($92.9K)** but second in profit per order at $31.60.
- Marketplace has the worst profit per order ($15.40) due to an $18.97 average platform fee per order.
- Social Commerce profit per order is $17.11, weighed down by $9.87 average platform fees.
- Both Website and Mobile App charge zero platform fees — all margin erosion in other channels is fee-driven.

CEO Insight	**Marketplace and Social Commerce platform fees consume 13-15% of margin. Negotiating fee caps or redirecting spend to owned channels (Website/App) would directly lift profitability.**

4. Return Rate Analysis (Q3)

Results by category

Category	Total Orders	Returned Orders	Revenue Lost	Return Rate	Rank
Electronics	267	23	$4,078	**8.61%**	1st
Books	239	20	$2,106	8.37%	2nd
Clothing	293	24	$3,209	8.19%	3rd
Sports	292	21	$2,008	7.19%	4th
Toys	257	18	$2,470	7.00%	5th
Home & Kitchen	200	12	$2,340	6.00%	6th
Beauty	205	12	$866	5.85%	7th
Food & Beverage	247	14	$3,505	**5.67%**	8th

Total revenue lost to returns across all categories: **$20,582.45**

Key findings

- Electronics has the highest return rate at 8.61% — the combination of high-value items and possible quality/fit issues suggests product description improvements could reduce this.
- Food & Beverage has the lowest return rate (5.67%) but the highest revenue lost in absolute terms ($3,505) due to high average order values.
- Clothing returns (8.19%) are expected given sizing uncertainty — a size guide or virtual try-on could reduce this meaningfully.
- Total refunds of $20,582 represent approximately 7.4% of gross revenue across the platform.

CEO Insight	**Electronics ($4,078 lost) and Clothing ($3,209 lost) are the highest-priority categories for return reduction initiatives. Each 1% reduction in Electronics return rate recovers approximately $180 in revenue.**

5. Marketing Platform ROAS Analysis (Q4)

Results

Platform	Total Spend	Revenue Attr.	Avg ROAS	Avg CPA	Spend %	Action
TikTok Ads	$57,229	$1,374,627	**24.4x**	$3.93	11.4%	**Grow**
Influencer	$97,663	$2,216,974	**23.4x**	$4.80	19.4%	**Grow**
Instagram Ads	$65,154	$1,024,639	17.0x	$5.55	12.9%	**Hold**
Google Ads	$152,546	$2,194,121	13.7x	$6.48	30.3%	**Reduce**
Facebook Ads	$106,452	$1,218,572	11.3x	$8.38	21.1%	**Reduce**
Email Mktg	$24,461	$117,681	**5.4x**	**$26.01**	4.9%	**Cut**

Key findings

- TikTok Ads delivers the best **ROAS at 24.4x — generating $24 of revenue per $1 spent, yet consumes only 11.4% of total budget.**
- Influencer marketing delivers 23.4x ROAS and 27.2% of total attributed revenue — the single largest revenue contributor.
- Google Ads consumes the largest budget share (30.3%) but only delivers 13.7x ROAS — disproportionate spend for the return.
- **Email Marketing is a clear underperformer:** 5.4x ROAS, $26 CPA (highest by far), and only 1.4% revenue share despite 4.9% of total spend.
- Facebook Ads is declining year-over-year — ROAS dropped from 12.6x in 2024 to 9.9x in 2025.

CEO Insight	**Email Marketing has a CPA of $26 — more than 6x the $3.93 CPA of TikTok. Budget reallocation from Email to TikTok would generate approximately 4.5x more conversions per dollar.**

6. Marketing Budget Cut Recommendation (Q5)

Context

Total marketing spend: $503,506. A 20% reduction requires saving $100,625 while minimising impact on revenue. The strategy targets the three lowest-ROAS platforms and focuses cuts within their worst-performing months.

Recommended cut plan

Platform	Total Spend	Avg ROAS	Cut %	Amount Cut	Action
Email Mktg	$24,461	5.4x	80%	$19,569	CUT — worst ROAS
Facebook Ads	$106,452	11.3x	30%	$31,936	CUT — declining
Google Ads	$152,546	13.7x	32%	$49,120	CUT — over-indexed
Instagram Ads	$65,154	17.0x	0%	—	HOLD
Influencer	$97,663	23.4x	0%	—	GROW
TikTok Ads	$57,229	24.4x	0%	—	GROW
TOTAL	**$503,506**		**20%**	**$100,625**	

Month-level cuts within Google & Facebook

Within Google Ads and Facebook Ads, cuts should be concentrated in months where ROAS fell below 5x:

Platform	Month	ROAS	Priority
Facebook Ads	Sep 2025	**2.51x**	Cut first
Facebook Ads	Aug 2024	**3.09x**	Cut first
Google Ads	Mar 2025	**3.32x**	Cut first
Google Ads	Sep 2025	**3.84x**	Cut first
Google Ads	Feb 2025	**3.89x**	Cut first
Facebook Ads	Mar 2025	**4.60x**	Reduce 50%
Google Ads	Jul 2025	**4.27x**	Reduce 50%

Key findings

- Email Marketing should be cut by 80% immediately — ROAS of 5.4x and CPA of $26 cannot be justified against any alternative.

- Facebook's ROAS declined from 12.6x (2024) to 9.9x (2025) — a downward trend that warrants a 30% budget reduction.
- Google Ads is over-indexed at 30.3% of budget but delivers only 13.7x ROAS. A 32% cut still leaves it funded for high-ROAS months.
- TikTok and Influencer are both trending upward — any freed budget should be reallocated to these platforms.
- The seasonal pattern shows Jul–Aug and Nov are consistently weak ROAS months — cuts in these periods minimise revenue impact.

CEO Recommendation	**Cut Email (80%), Facebook (30%), Google (32%) targeting worst months. Reallocate $50K+ of savings to TikTok and Influencer for an estimated net ROAS improvement of 3-5x on reinvested spend.**